# PII Masking via Fine-Tuned BERT and Zero-Shot LLMs

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`bert-base-cased` fine-tuned on WikiNeural + synthetic emails — Zero-shot evaluation: LLaMA-3.1-8B and GPT-4o

## 1 Approach & Data Preparation

The task required detecting and masking two PII categories **person names** and **email addresses** using a fine-tuned encoder transformer and zero-shot LLM prompting. The WikiNeural dataset provided rich name annotations but contained no email addresses, requiring a principled injection strategy before any model could be trained or evaluated.

### 1.1 Two Test Sets: Synthetic and Independent

Two distinct evaluation sets were constructed, each testing a different aspect of generalisability.

**Synthetic test set.** The provided `test_data.json` (3,650 sentences) formed the base. It contains person name annotations from WikiNeural but no emails, so synthetic email addresses using completely unseen institutional domains such as `uni.edu.pk`, `mail.gov.uk`, `work.io`, `dept.org` to prevent domain string memorisation. This set evaluates in distribution name recall combined with out of domain email generalisation.

**Independent test set.** The full WikiNeural English test split was loaded and all sentences already present in the provided training data were removed via exact token sequence matching. The 2,500 remaining unique sentences (1,116 PER entities) form a set the model has never encountered. Person names in this split were supplemented using the **Faker** library to introduce synthetic but realistic names, and emails were again injected with unseen institutional domains. This set is the true measure of out of distribution performance.

### 1.2 Realistic Email Injection

A naive uniform injection would produce an unnatural distribution. Real world email prevalence in free text is approximately 8%, but that density would make the training signal too sparse to learn from. A **30% injection probability** was chosen as a practical compromise dense enough to train on, still reflecting the minority nature of the label.

The local part of each injected email was derived from the adjacent name, matching four naturally occurring patterns: `first.last{N}`, `first{N}`, `last_suffix`, and random alphanumeric.

To prevent the model from learning a positional shortcut, emails were not inserted exclusively after a PER span. Instead, **50% of injections were placed immediately after the nearest PER span**, while the remaining **50% were inserted at a randomly chosen position anywhere in the token sequence**, regardless of proximity to a name. This split ensures the model cannot rely on positional co-occurrence with a person entity as a proxy for email detection; it must instead learn the structural signature of the email itself.

The most consequential design decision was splitting every email into its **atomic sub tokens**: local part, `@`, each domain segment, dots, and the domain, all tagged under a single `B-EMAIL`/`I-EMAIL` BIO chain. This granularity serves two purposes simultaneously. First, it forces the model to learn the internal structure of an email address as a sequential pattern anchored by the `@` symbol, rather than treating the address as an opaque string. Second, the fine grained tagging produces a healthy number of `I-EMAIL` training tokens relative to `B-EMAIL` tokens, giving the model sufficient signal to learn continuation spans and preventing the collapsed single token labelling observed in zero-shot LLMs. After injection the training corpus contained 8,508 email entities across 28,516 sentences.

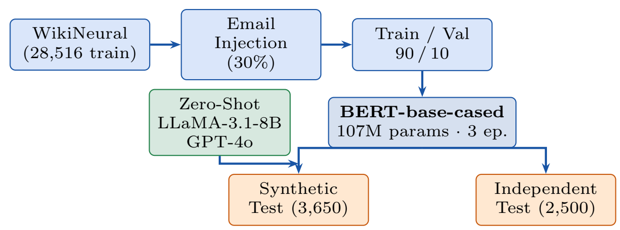


**Figure 1:** Pipeline: name-derived email injection, BERT fine-tuning, and evaluation on two structurally distinct test sets with domain-shifted emails.

## 2 Model Training

**bert-base-cased** was chosen because capitalisation is a strong lexical cue for person names; an uncased model would discard this signal. The classification head maps to five BIO labels {`O, B-PER, I-PER, B-EMAIL, I-EMAIL`}. Subword continuation tokens are masked (`label = -100`) so loss is computed only on the first subword of each word.

Training used 3 epochs on a Tesla T4 GPU (batch 32, lr $2 \times 10^{-5}$, weight decay 0.01, 200 warmup steps, FP16). Best checkpoint selected on validation F1; training loss converged to 0.038.

## 3 Evaluation Results

### 3.1 Fine-Tuned BERT

**Table 1:** BERT span-level results on both test sets.

| Test Set | Entity | Acc | FPR | FNR | P | R |
|---|---|---|---|---|---|---|
| Indep. | EMAIL | 0.994 | 0.006 | 0.007 | 0.94 | 0.98 |
| | PER | | | | 0.78 | 0.98 |
| Synth. | EMAIL | 0.997 | 0.003 | 0.007 | 0.93 | 0.98 |
| | PER | | | | 0.97 | 0.98 |

Email F1 holds at **0.96** across both test sets despite entirely different domain strings at inference confirming the model learned structural email patterns rather than surface domain strings. Person name precision is lower on the independent set (0.78 vs 0.97) because Wikipedia names outside the training distribution include more ambiguous, short, or culturally diverse tokens that BERT conflates with common words.

### 3.2 Zero-Shot LLMs

Both LLaMA-3.1-8B and GPT-4o were evaluated on 250 samples from the synthetic test set without any fine tuning.

A structured system prompt was crafted specifying BIO tagging rules. email sub token labelling.

**Table 2:** Per-entity zero-shot results on 250 synthetic test samples.

| Model | Entity | Acc | FPR | FNR | P | R |
|---|---|---|---|---|---|---|
| LLaMA-3.1-8B (Zero-Shot) | EMAIL | 0.898 | 0.052 | 0.268 | 0.516 | 0.474 |
| | PER | | | | 0.602 | 0.643 |
| GPT-4o (Zero-Shot) | EMAIL | 0.957 | 0.033 | 0.066 | 0.940 | 0.928 |
| | PER | | | | 0.723 | 0.867 |

**Table 3:** Overall comparison across all three approaches.

| Model | Acc | Macro F1 | FPR | FNR |
|---|---|---|---|---|
| BERT (fine-tuned) | **0.997** | **0.974** | **0.003** | 0.007 |
| GPT-4o (0-shot) | 0.957 | 0.861 | 0.033 | 0.066 |
| LLaMA-3.1-8B (0-shot) | 0.898 | 0.554 | 0.052 | 0.268 |

GPT-4o reaches a macro F1 of **0.86** with no supervision, achieving near-parity with BERT on email detection (B-EMAIL F1: 0.89). LLaMA-3.1-8B, however, collapses on email boundary detection (B-EMAIL F1: 0.31) and its FNR of 0.27 means it misses more than one in four PII tokens unacceptable for a compliance pipeline.

## 4 Error Analysis

Analysis of 176 token level misclassifications on the independent test set revealed two dominant failure modes.

**Context bleed (O → PER), 93 cases.** Common words adjacent to a person name *"the"*, *"of"*, *"city"* were incorrectly pulled into the PER span. BERT's attention picks up residual signal from neighbouring name tokens when surrounding context is sparse.

**BIO boundary confusion (I-PER → B-PER), 16 cases.** Continuation tokens of multi word names are sometimes labelled as new span beginnings.

For the LLMs, the primary failure was **email token granularity**. LLaMA consistently tagged the full email string as a single `B-EMAIL` token rather than distributing BIO labels across sub-tokens (`@`, domain parts). This explains the high I-EMAIL recall gap (0.61) relative to GPT-4o (0.96), which adhered to the sub token anchoring rule more reliably.

## 5 Improvement Suggestions

**Negative augmentation.** Adding sentences with email like strings that are not PII product codes, URLs, ISBNs would tighten the decision boundary and reduce false positives on words adjacent to name spans.

**LLM structured output.** For LLaMA, few-shot examples inside the prompt showing correct BIO sub token labelling of emails improve boundary compliance significantly.

## 6 Key Insights

Three findings stand out. First, the deliberate domain shift between training and test email sets consumer services in training, institutional domains at inference proved the model learned structural email patterns rather than surface strings; email F1 held at 0.96 under full distribution shift of domain names. Second, the three-model comparison shows that fine-tuning a 107M parameter model on task specific data still outperforms a much larger frontier model in zero-shot mode, while a small open weights model is not competitive for production use without fine-tuning. Third, LLaMA's FNR of 0.27 is a security risk: missing roughly one in four PII tokens defeats the purpose of automated masking. For any compliance critical deployment, supervised fine tuning is not optional it is necessary.

**Reproducibility.** Dataset: `Babelscape/wikineural` (HuggingFace). Model: `bert-base-cased`. LLM APIs: Groq (LLaMA-3.1-8B-Instant) and OpenAI (GPT-4o). Random seeds fixed at 42. All code in `Final_code.ipynb`. Code repository: `github.com/AwaisGill143/Masking-PII`.